

office hours 12:30 - 14:00 Monday

grade: midterm presence homework final term.
30% 10% 20% 40%

midterm: Apr. 13th Monday. 8th week.

Recall: Representation of quiver: $V_i, i \in Q_0, \varphi_\alpha, \alpha \in Q,$

$$V_{s(\alpha)} \xrightarrow{\varphi_\alpha} V_{t(\alpha)}, \quad \varphi_\alpha \in \text{Hom}(V_{s(\alpha)}, V_{t(\alpha)})$$

$$M = (V_i, \varphi_\alpha)_{i \in Q_0, \alpha \in Q},$$

Example: $\begin{array}{ccc} \bullet & \xrightarrow{\quad} & \bullet \\ 1 & & 2 \end{array}$

Consider representation $M: \begin{array}{ccc} k & \xrightarrow[\begin{bmatrix} 0 \\ 1 \end{bmatrix}]{\begin{bmatrix} 1 \\ 0 \end{bmatrix}} & k^2 \end{array} \quad M = (V_i, \varphi_\alpha)$

$N: \begin{array}{ccc} k^2 & \xrightarrow[\begin{bmatrix} 1 \\ 1 \end{bmatrix}]{\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}} & k^2 \end{array} \quad N = (U_i, \psi_\alpha)$

Consider $f: M \rightarrow N$ morphism from M to N

$$f = (f_i)_{i \in Q_0}, \quad f_i \in \text{Hom}(V_i, U_i)$$

for each $\alpha: i \rightarrow j$ in Q , the following diagram commutes:

$$\begin{array}{ccc} V_i & \xrightarrow{\varphi_\alpha} & V_j \\ f_i \downarrow & & \downarrow f_j \\ U_i & \xrightarrow{\psi_\alpha} & U_j \end{array}$$

We use $\text{Rep } Q$ denote the category of representations of Q

For our example : $f: M \rightarrow N$, $f = (f_1, f_2)$

$$\begin{array}{ccc}
 k & \xrightarrow[\begin{smallmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \end{smallmatrix}]{\begin{smallmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \end{smallmatrix}} & k^2 \\
 \downarrow f_1 = \begin{bmatrix} a \\ b \end{bmatrix} & & \downarrow \begin{bmatrix} c & d \\ e & f \end{bmatrix} = f_2 \\
 k^2 & \xrightarrow[\begin{smallmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \end{smallmatrix}]{\begin{smallmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \end{smallmatrix}} & k^2
 \end{array}$$

$$\begin{bmatrix} c & d \\ e & f \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} c & d \\ e & f \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

check : $\begin{bmatrix} c \\ e \end{bmatrix} = \begin{bmatrix} a+b \\ b \end{bmatrix}$

$$\begin{bmatrix} a+b \\ a+b \end{bmatrix} = \begin{bmatrix} d \\ f \end{bmatrix}$$

$$\begin{cases} c = d = f = a+b \\ e = b \end{cases}$$

Hence. $f = (\begin{bmatrix} a \\ b \end{bmatrix}, \begin{bmatrix} a+b & a+b \\ b & a+b \end{bmatrix})$, $\dim \text{Hom}(M, N) = 2$

Direct sum and indecomposable representations

If M and N are representations of Q

then $M \oplus N$ is a representation with vector spaces $V_i + U_i, \psi_1 + \psi_2$

$$M \oplus N = (V_1 + U_1, \begin{bmatrix} \psi_1 \\ \psi_2 \end{bmatrix})$$

Inductively we can consider $M_1 \oplus \dots \oplus M_k$

If N is a representation, can it be decomposed into direct sum of two non-zero representations?

Remark: In our class we may face with indecomposable $\neq$ irreducible, which is a bit different from group representation course.

Example: $1 \longrightarrow 2 \longleftarrow 3$

consider representation: $M: k \xrightarrow{1} k \xleftarrow{0} 0$

$$N: k^2 \xrightarrow{\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}} k^2 \xleftarrow{\begin{bmatrix} 1 \\ 1 \end{bmatrix}} k$$

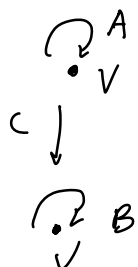
$$M \oplus N: k \oplus k^2 \xrightarrow{\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}} k \oplus k^2 \xleftarrow{\begin{bmatrix} 0 & 1 \\ 1 & 1 \end{bmatrix}} 0 \oplus k$$

indeed: $k^3 \xrightarrow{\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}} k^3 \xleftarrow{\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}} k$

Definition: We call M indecomposable if it can not be written as a direct sum of two non-zero representations of Q

Example:  A, B, C are $n \times n$ maps

C is an isomorphism



$$\begin{array}{ccc} V & \xrightarrow{A} & V \\ C \downarrow & & \downarrow C \\ V & \xrightarrow{B} & V \end{array}$$

We have $A = C^{-1}BC$

Here the isomorphism classes is corresponding to JNF

To describe $\text{Rep } Q$ we will need to describe all isomorphism classes of indecomposable representations.

Krull-Schmidt theorem: Any $M \in \text{Rep } Q$, (finite)

can be decomposed into finite number of indecomposables and its decomposition is unique up to order

kernel and cokernel representations:

$$f: V \rightarrow U \quad \ker f = \{ v \in V \mid f(v) = 0 \}, \quad \text{Im } f = \{ u \mid \exists v \in V, f(v) = u \}$$

$$\text{coker } f = U / \text{Im } f$$

Let $f: M \rightarrow N$ be morphism from M to N , $M, N \in \text{Rep } Q$, $M = (U_i, \varphi_\alpha)$, $N = (U_i, \psi_\alpha)$

Define $\ker f \in \text{Rep } Q$ as (L_i, χ_α) where $L_i = \ker f_i$

$$\chi_\alpha = \varphi_\alpha|_{L_i}$$

$$\chi_\alpha(v_i) = \varphi_\alpha(v_i), \quad \forall v_i \in L_i$$

We want to show that $\ker f$ indeed in $\text{Rep } Q$. i.e.: $\chi_\alpha: L_i \rightarrow L_j$

$\Rightarrow$ to show that $\forall v_i \in L_i, \chi_\alpha(v_i) \in L_j, \alpha: i \rightarrow j$

$$\Rightarrow \forall v_i \in L_i, f_i(v_i) = 0$$

$$f_j(\chi_\alpha(v_i)) = f_j(\varphi_\alpha(v_i)) = \varphi_\alpha(f_i(v_i)) = 0$$

$$\Rightarrow \chi_\alpha(v_i) \in \ker f_j = L_j \quad \square$$

Then inclusion: $L_i \hookrightarrow V_i$ defines injective morphism from $\ker f$ to M ,

$$\begin{array}{ccc} L_i & \xrightarrow{\alpha} & L_j \\ \downarrow & & \downarrow \\ M_i & \xrightarrow{\varphi_\alpha} & M_j \end{array}$$

$\text{Coker } f: (N_i / \text{Im}(f_i), \gamma_\alpha), \alpha: i \rightarrow j$

where $\gamma_\alpha (n_i + f_i(M_i)) := \psi_\alpha(n_i) + f_j(M_j) \quad N_i / \text{Im}(f_i) \rightarrow N_j / \text{Im}(f_j)$

We need to show that γ_α is well-defined.

Let $n_i + f_i(M_i) = n'_i + f_i(M_i)$ then $n_i - n'_i \in f_i(M_i)$
 $\Rightarrow \exists m_i, f_i(m_i) = n_i - n'_i$

$$\begin{aligned} \psi_\alpha(n_i - n'_i) &= \psi_\alpha(f_i(m_i)) \\ &= f_j(\varphi_\alpha(m_i)) \in \text{Im } f_j = f_j(M_j) \end{aligned}$$

$$\Rightarrow \psi_\alpha(n_i) - \psi_\alpha(n'_i) \in f_j(M_j)$$

Consider the projection $N \rightarrow \text{Coker } f$, it is a surjective morphism in $\text{Rep } Q$. $\text{proj}: N_i \rightarrow N_i / \text{Im } f_i$

Equivalent definition of $\ker f$

Let $f: M \rightarrow N$ be morphism, we call $g: L \rightarrow M$ the kernel representation of

f if $fg = 0$ and for any $h: X \rightarrow M$ with $fh = 0$

there exist $u: X \rightarrow L$ s.t. $h = gu$. i.e the following unique

diagram commutes:

